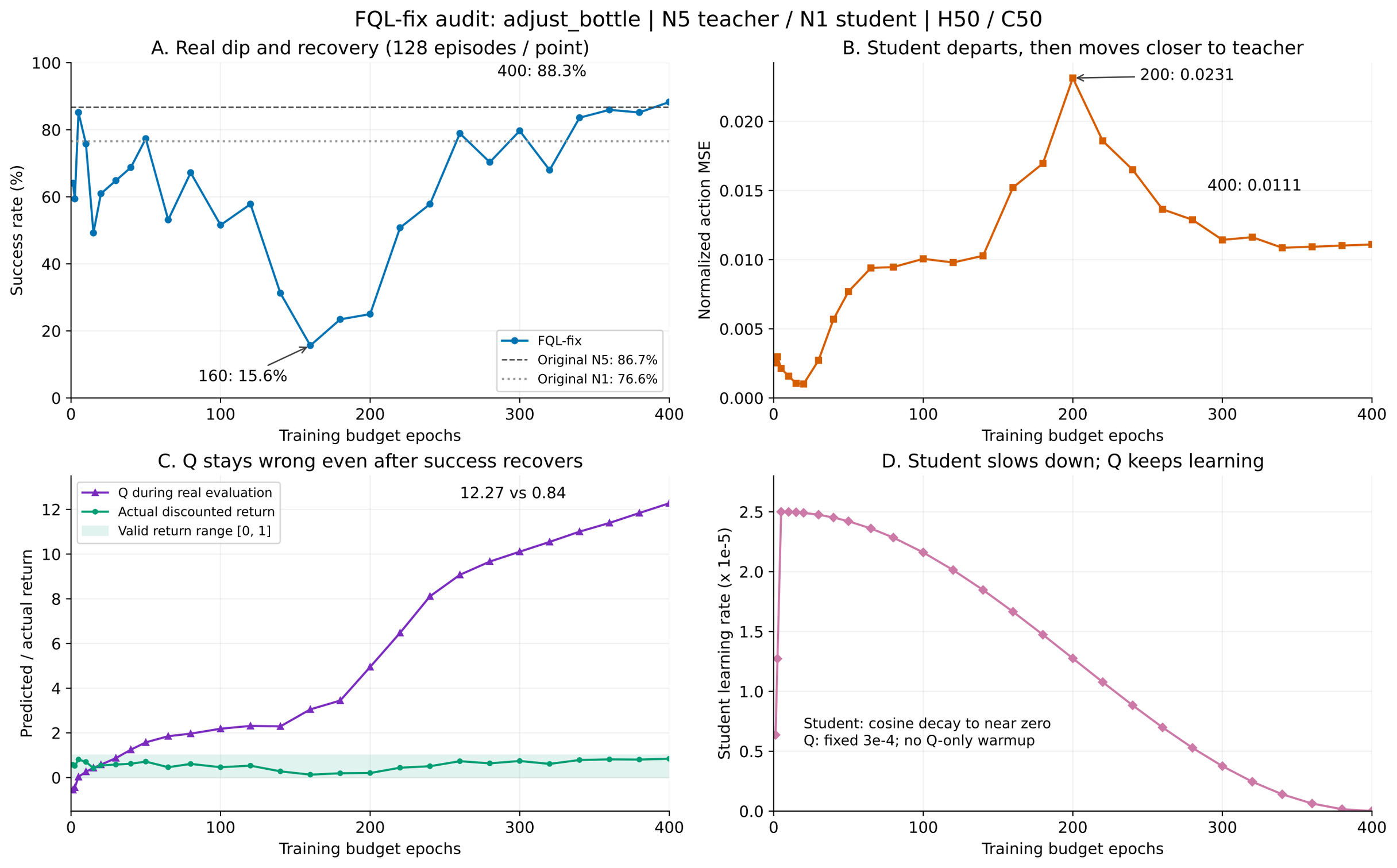
FQL-fix audit: adjust_bottle | N5 teacher / N1 student | H50 / C50
A. Real dip and recovery (128 episodes / point)
Success rate (%)
Training budget epochs
400: 88.3%
160: 15.6%
FQL-fix
Original N5: 86.7%
Original N1: 76.6%
B. Student departs, then moves closer to teacher
Normalized action MSE
Training budget epochs
200: 0.0231
400: 0.0111
C. Q stays wrong even after success recovers
Predicted / actual return
Training budget epochs
Q during real evaluation
Actual discounted return
Valid return range [0, 1]
12.27 vs 0.84
D. Student slows down; Q keeps learning
Student learning rate (x 1e-5)
Training budget epochs
Student: cosine decay to near zero
Q: fixed 3e-4; no Q-only warmup